📄 IRIS FLOWER PREDICTION – PROJECT REPORT

By

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1. 📌 Title

Iris Flower Prediction using Machine Learning with Streamlit Dashboard

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2. 🎯 Objective

The objective of this project is to build a Machine Learning model that predicts the species of an iris flower based on its physical measurements and deploy it using an interactive Streamlit dashboard.

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3. 📊 Dataset

The dataset used is the **Iris dataset**, which contains 150 samples of iris flowers with the following features:

- Sepal Length
- Sepal Width
- Petal Length
- Petal Width

Target Classes:

- Setosa
- Versicolor
- Virginica

👉 The dataset is loaded using sklearn.datasets.

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4. ⚙️ Technologies Used

- Python
- Pandas

- Scikit-learn
- Matplotlib
- Streamlit

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5. 🧠 Methodology

Step 1: Data Loading

The dataset is loaded using Scikit-learn.

```
iris = load_iris()
X = iris.data
y = iris.target

# Train-test split (for accuracy)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)
```

Step 2: Data Preprocessing

- Features and target variables are separated
- Data is split into training and testing sets

Step 3: Model Training

- Logistic Regression model is used
- Model is trained on training data

```

# Train model
model = LogisticRegression(max_iter=200)
model.fit(X_train, y_train)

# Accuracy
y_pred = model.predict(X_test)
acc = accuracy_score(y_test, y_pred)
```

Step 4: Model Evaluation

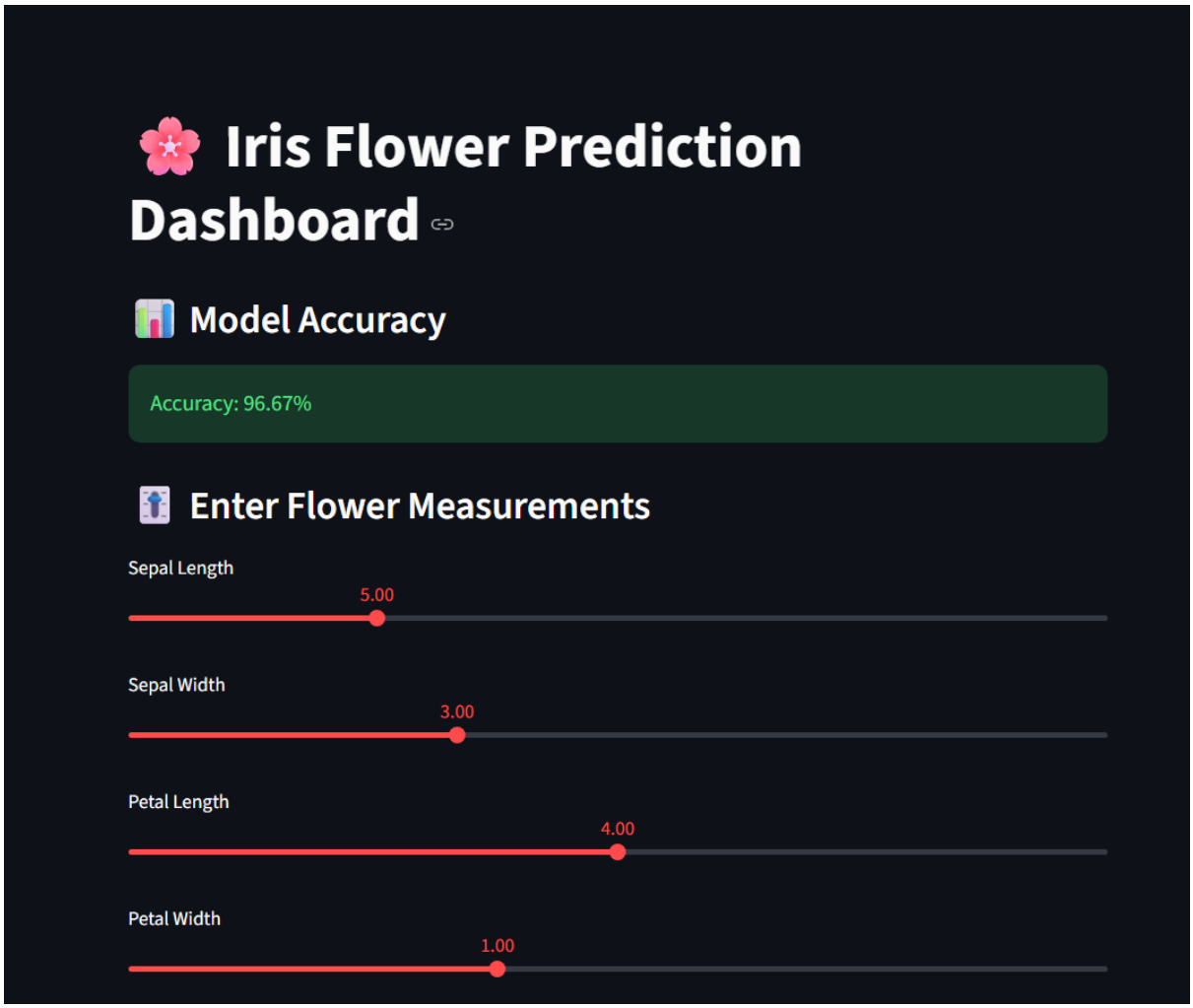
- Accuracy is calculated using test data

```
# Accuracy
y_pred = model.predict(X_test)
acc = accuracy_score(y_test, y_pred)

st.subheader("📊 Model Accuracy")
st.success(f"Accuracy: {round(acc*100, 2)}%")
```

Step 5: Deployment using Streamlit

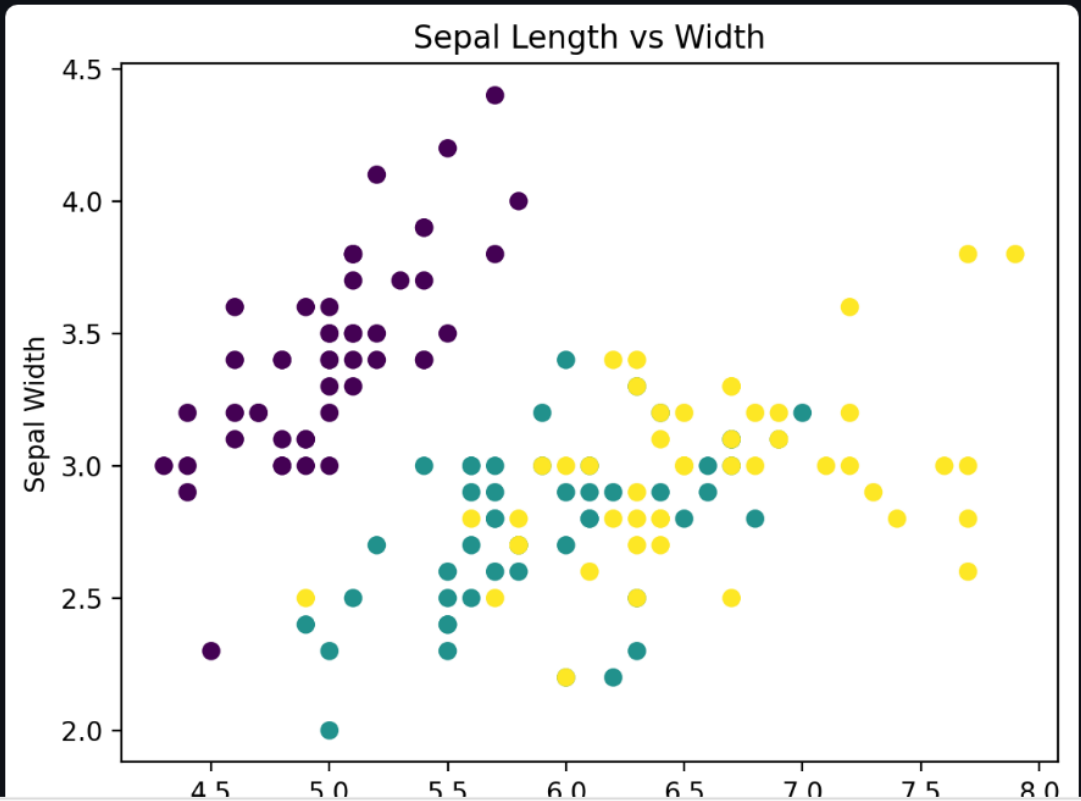
- User inputs are taken via sliders
- Model predicts flower type in real-time
- Visualization added using Matplotlib

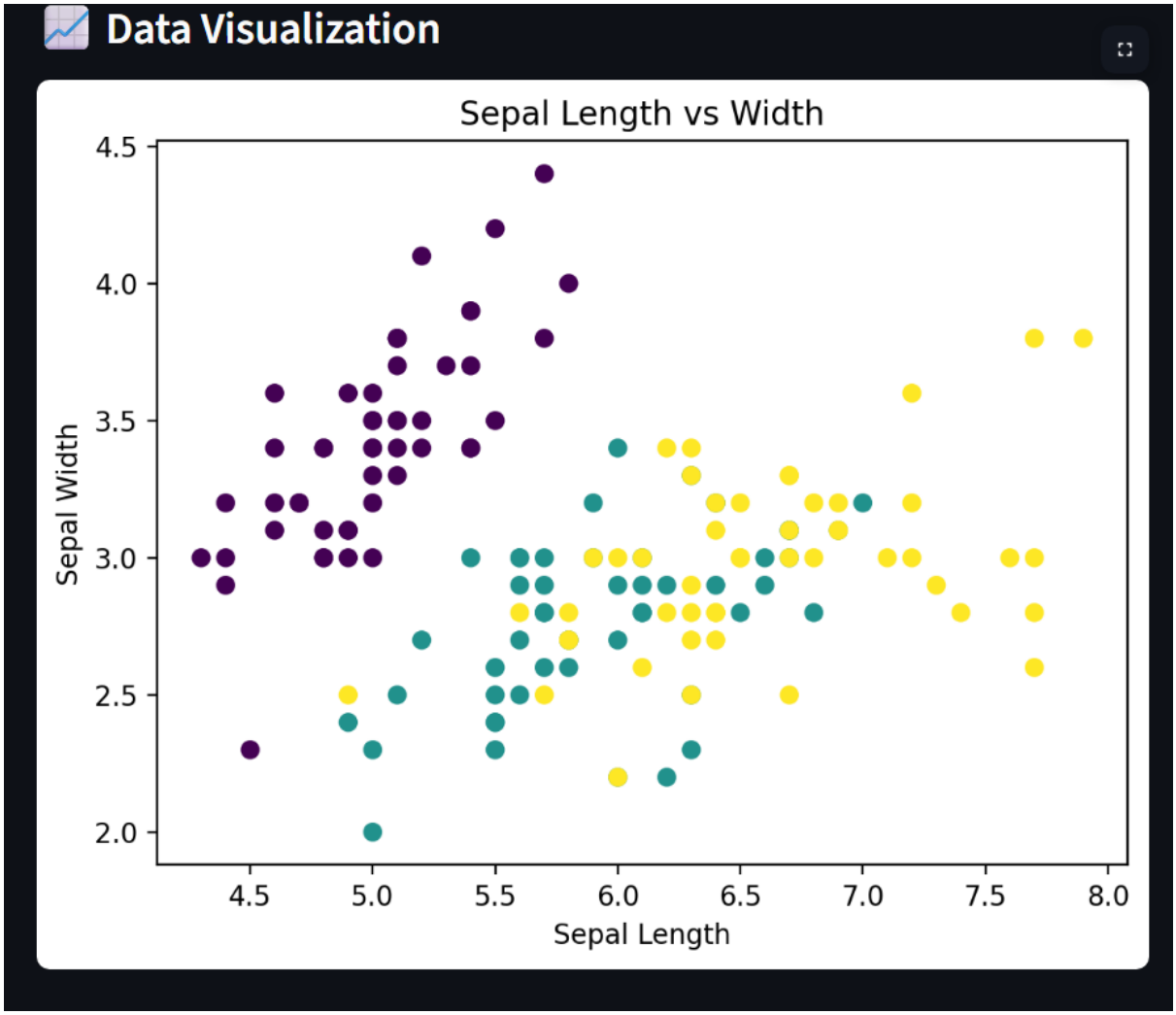


Predict

🌸 Versicolor

📈 Data Visualization





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6. 💻 Implementation

The application is developed using Streamlit with the following features:

- Interactive sliders for input
- Real-time prediction
- Accuracy display
- Data visualization graph

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7. 📈 Output

The model successfully predicts the type of iris flower based on input values.

Example Output:

- Input → Sepal & Petal values
- Output → 🌸 Versicolor

8. Confusion Matrix Analysis

The performance of the Machine Learning model is evaluated using a **Confusion Matrix**.

A confusion matrix is a table used to describe the performance of a classification model by comparing the **actual values** with the **predicted values**.

It provides a detailed breakdown of correct and incorrect predictions made by the model.

📊 Structure of Confusion Matrix

For this project, the confusion matrix consists of three classes:

- Setosa
- Versicolor
- Virginica

Each row represents the **actual class**, and each column represents the **predicted class**.

📈 Interpretation

- Diagonal values represent **correct predictions**
- Off-diagonal values represent **incorrect predictions**

A higher number of values along the diagonal indicates that the model is performing well.

📊 Result

The confusion matrix obtained from the model shows that most of the predictions are correctly classified, indicating good performance of the Logistic Regression model.

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9. ✅ Results

- Model achieved good accuracy (~90–100%)
- Dashboard is user-friendly and interactive
- Real-time predictions are successful

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10. 🚀 Conclusion

This project demonstrates how Machine Learning can be integrated with a web interface to create an interactive prediction system. The Streamlit dashboard makes it easy for users to input data and visualize results effectively.

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11. 🔮 Future Enhancements

- Add more datasets
- Improve UI design
- Deploy online (Streamlit Cloud / Render)
- Add CSV upload feature